AE 358 RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

LECTURE 2

RENEWABLE ENERGY SYSTEMS AND THEIR POTENTIAL

Lecturer: Francis Kemausuor

COURSE STUDY GUIDE

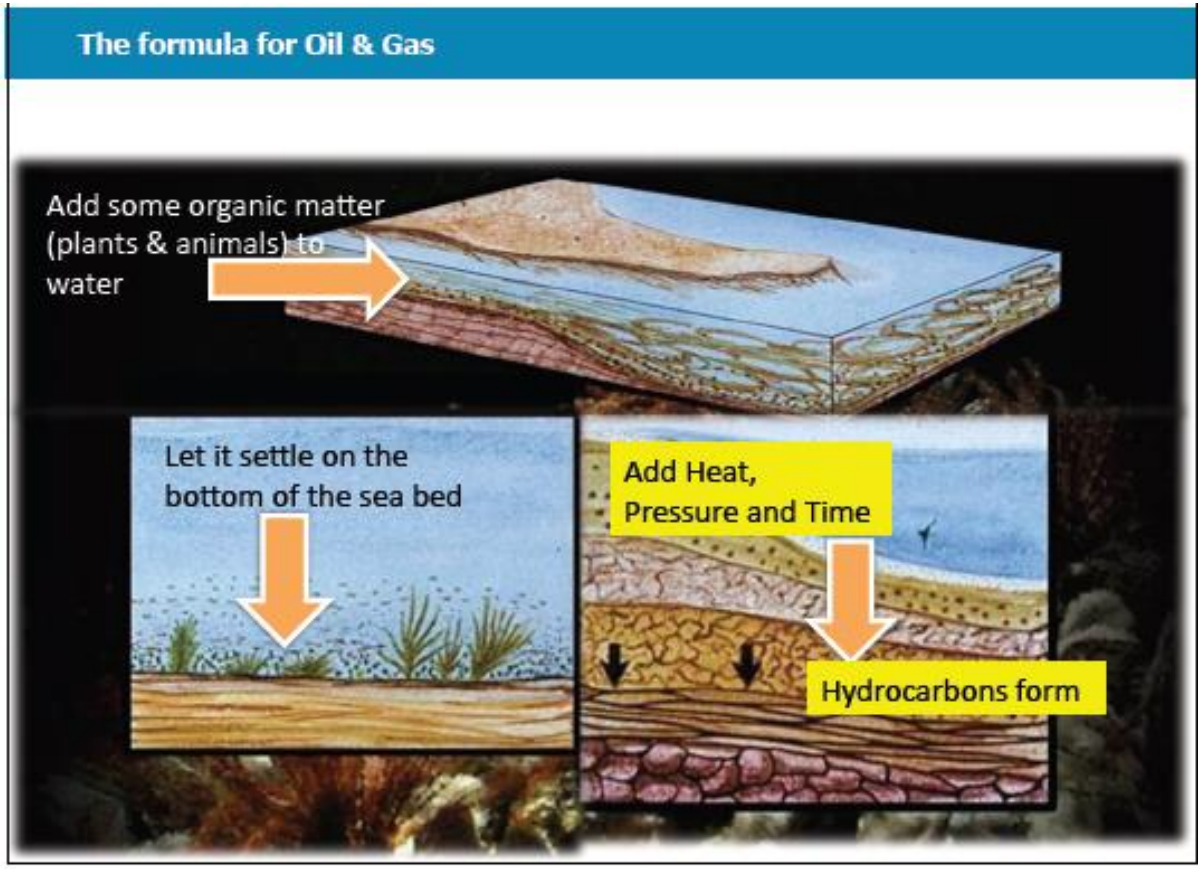
Date	Meeting/ Week #	Topic for the day	Activity
14-16 Jan	1		Lecture
19-23 Jan	2	Introduction to Energy Systems	Lecture
26-30 Jan			
2-6 Feb	3	Renewable Energy Systems and their Potential	Lecture
9-13 Feb	4	Hydropower	Lecture
16-20 Feb	5	Bioethanol / Group assignments	Lecture
23-27 Feb	6	Biodiesel	Lecture
2-6 Mar	7	Biogas	Lecture
9-13 Mar	8	Mid-semester exams	Quiz
16-20 Mar	9	Discuss group assignment challenges	Lecture
23-27 Mar	10	Solar Energy	Lecture
30 – 3 Apr	11	Wind Energy	Lecture
6-10 Apr	12	Renewable Energy Applications in Agriculture	Lecture
13-17 Apr	13	GROUP PRESENTATIONS	Student-led
13-17 April	14	GROUP PRESENTATIONS	Student-led
20-24 April	15	Revision	

Programme for today

- **Origin of fossil fuels**
  - Advantages and disadvantages of fossil fuel
- **Renewable Energy Sources**
  - Hydropower
  - Solar power
  - Wind power
  - Geothermal power
  - Tidal power
  - Biomass and biofuels

Formation of Oil and Gas

Fossil fuels are sources of energy that were formed by the death, decay, and transformation, or diagenesis, of life

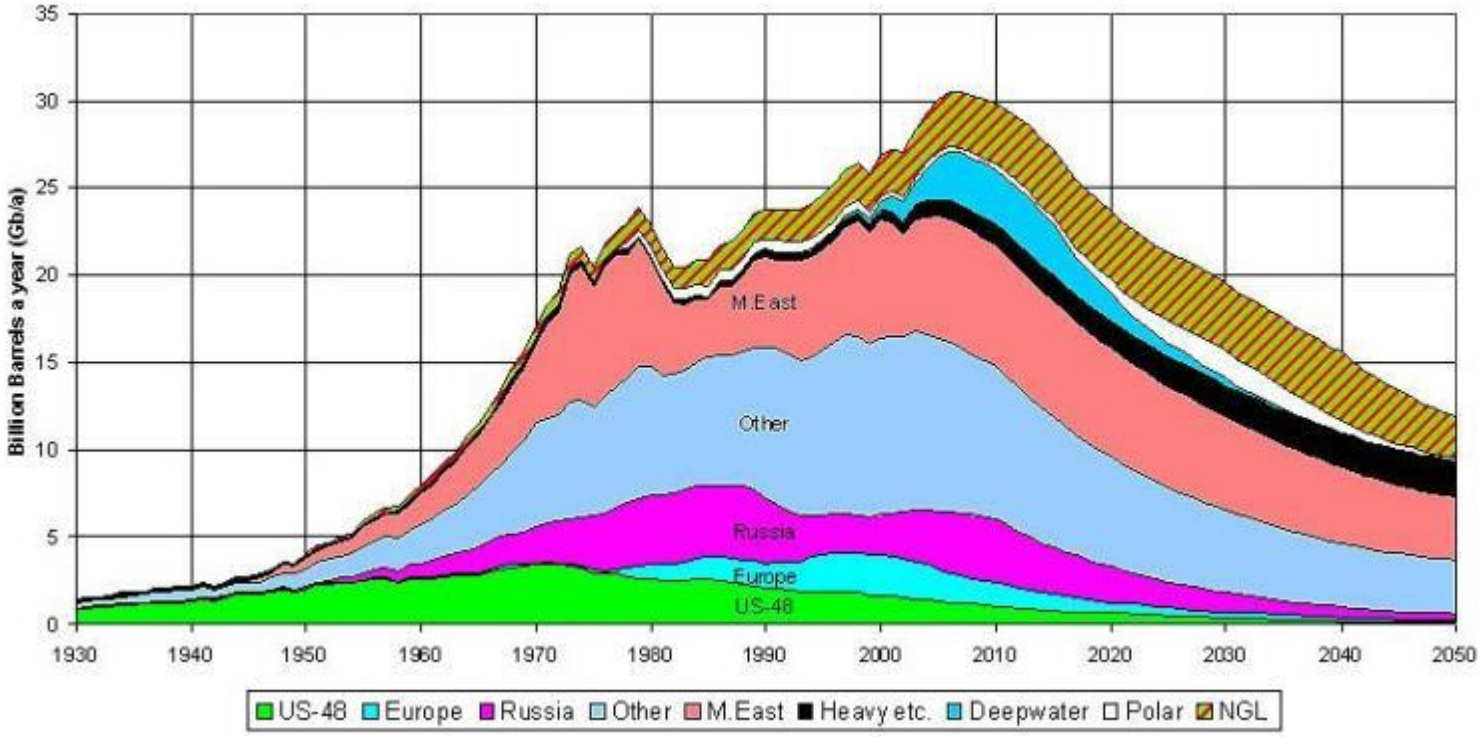


The Concept of Peak Oil

- The current global energy supply is highly dependent on fossil sources (crude oil, coal, natural gas).
- There is some disagreement among scientists on how long this fossil resource will last but according to researchers, the “peak oil production” has already occurred or it is expected to occur soon.

The Concept of Peak Oil

- The peak oil production is defined as “the point in time at which the maximum rate of global production of crude oil is reached, after which the rate of production enters its terminal decline”.

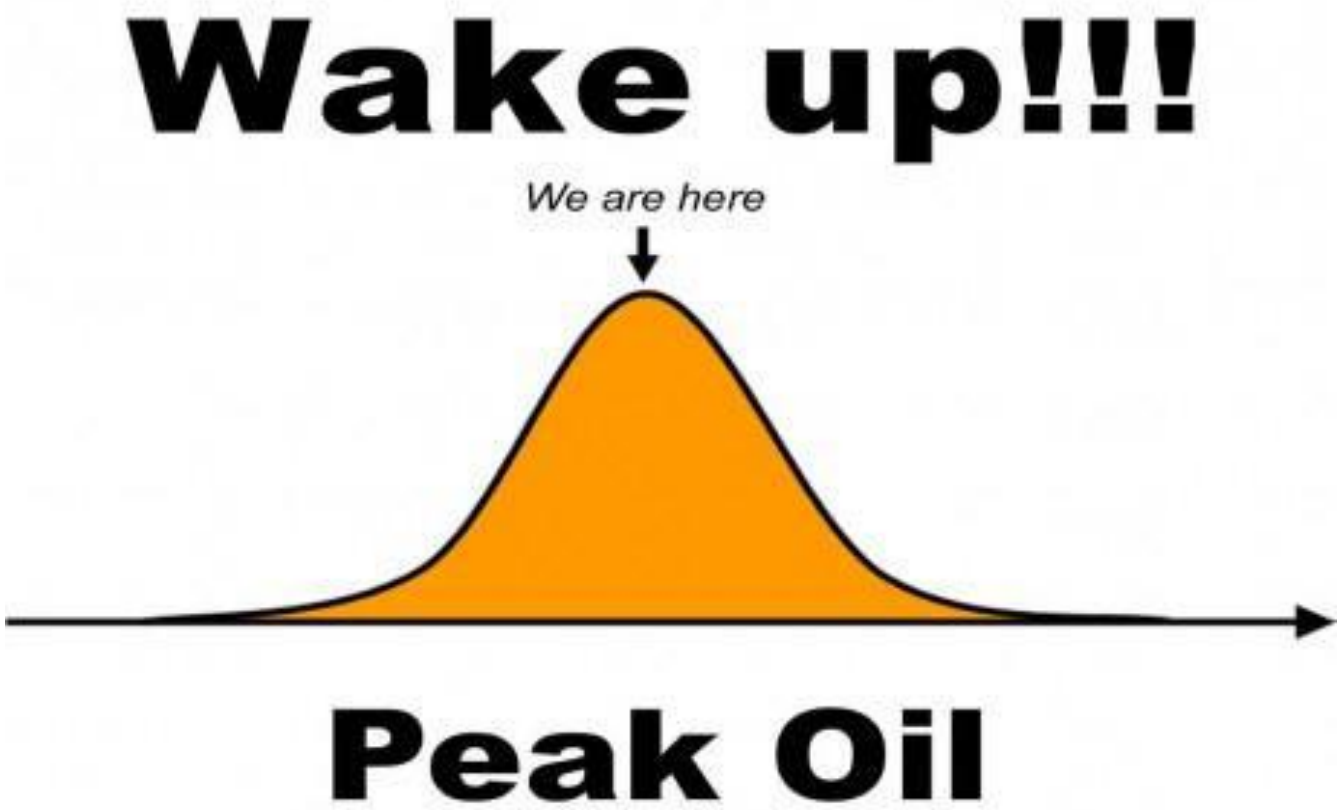


Advantages of fossil fuel

1) Capacity to generate huge amounts of electricity in just a single location.
2) Coal is in abundant supply and when used in power plants, they are very cost effective.
3) Transporting oil and gas to power stations can be made through the use of pipes making it an easy task.
4) Power stations that make use of fossil fuel can be constructed in almost any location, as long as large quantities of fuel can be easily brought to the power plants.

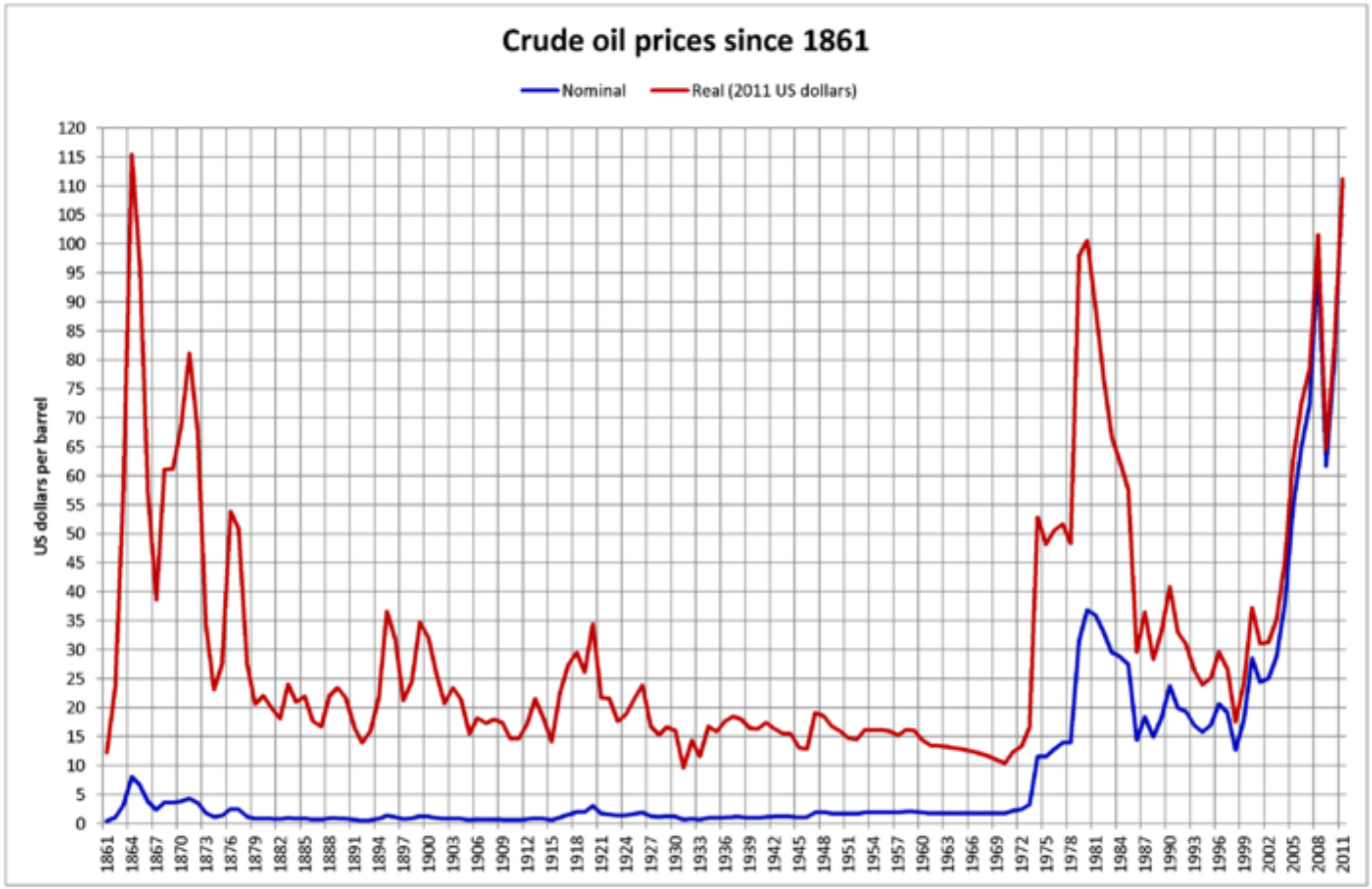
Disadvantages of fossil fuel

- oil is peaking…



Disadvantages of fossil fuel

- …and price is going up!!!



Disadvantages of fossil fuel

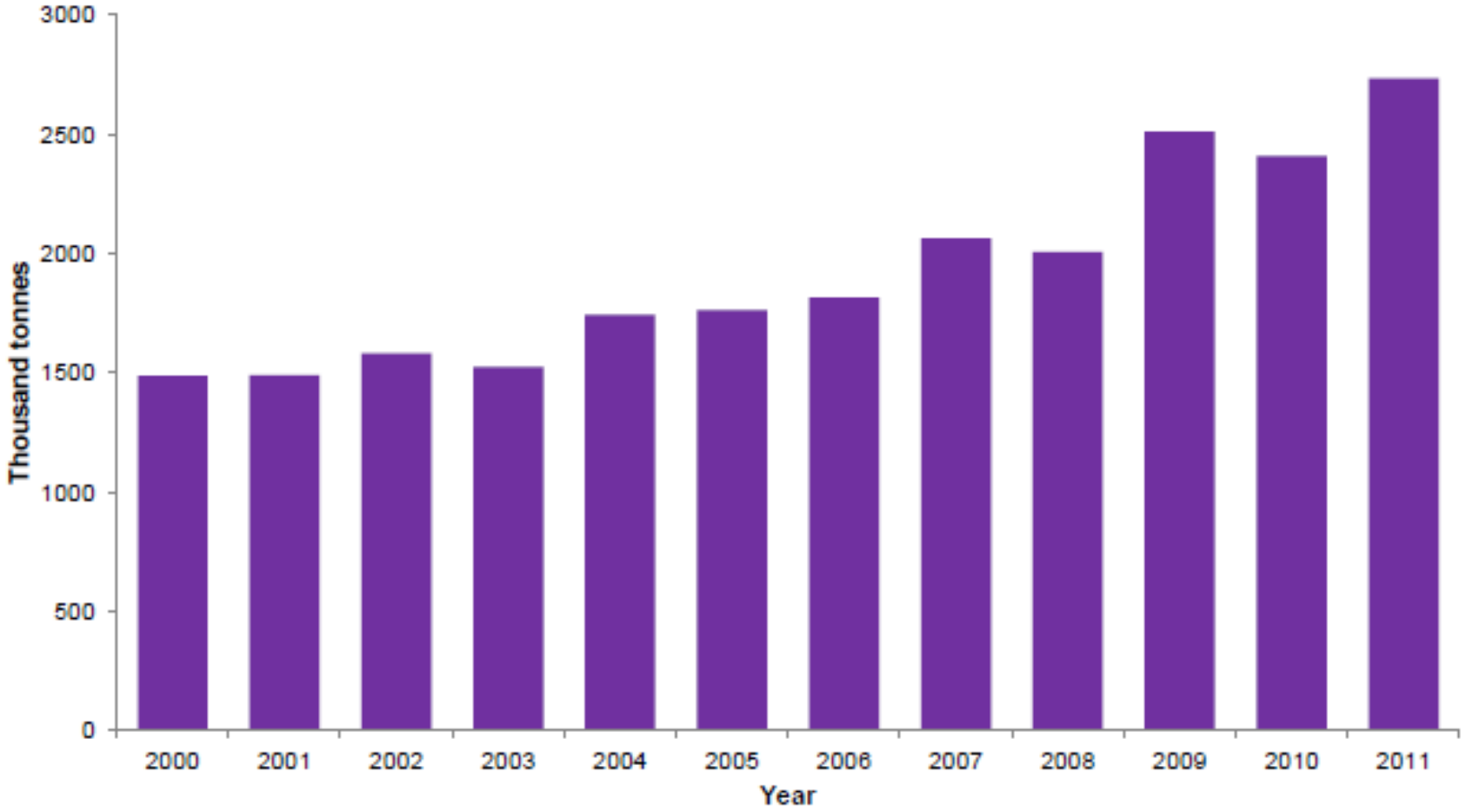
- Fossil oil also contributes to increased GHG levels



How much fossil fuel does the world consume? Brain teaser!!

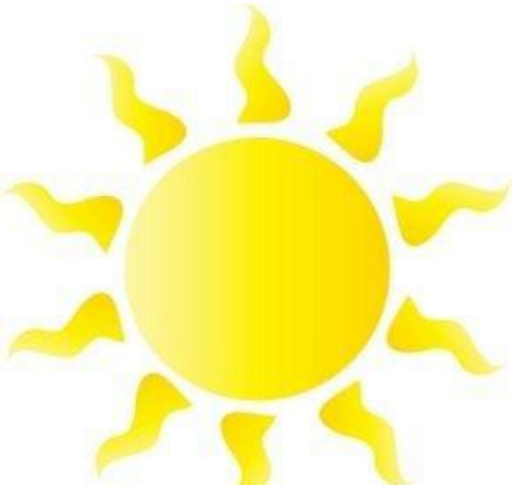
- How much oil per day?
  - About 90 million barrels!!!
- How much gas?
  - About 3,500 billion cubic meters
- How much coal?
  - About 8,000 million tonnes
- How much oil is Ghana producing?
  - About 110,000 barrels per day

Ghana petroleum consumption



RENEWABLE ENERGY SOURCES

Introduction





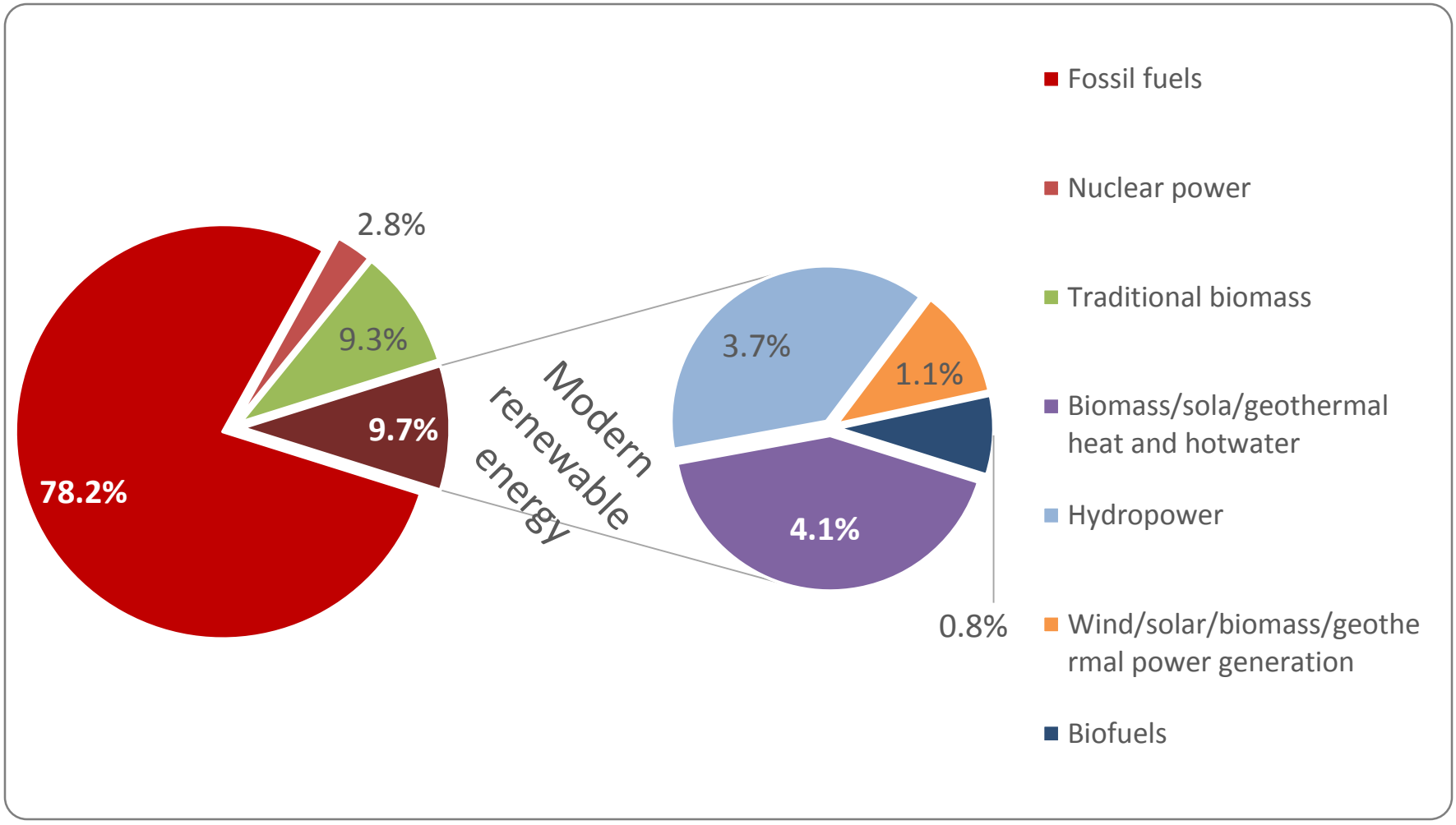


- Renewable energy is energy which comes from natural resources such as sunlight, biomass, wind, rain, tides, and geothermal heat, which are renewable (naturally replenished).

Introduction

- Renewable energy sources are fundamentally different from fossil fuels, and do not produce as many
- In view of the fact that fossil oil is peaking, renewable energy will play an important role in the future of our energy demand.
- While many renewable energy projects are large-scale, renewable technologies are also suited to rural and remote areas, where energy is often crucial in human development.

Global use of renewable energy (2011)



Global energy from RE sources? Assignment for the week!!!

1) How much energy is produced globally from solar?
2) How much energy is produced globally from wind?
3) How much energy is produced globally from hydropower?
4) How much energy is produced globally from traditional biomass?
5) Which countries are the leaders in each of the above named technologies? (Name the first two)
6) How much capacity was added in 2013 for the leading countries and also globally?

- **Reference: RENEWABLES 2014 GLOBAL STATUS REPORT (free online). See link below:**
- http://www.ren21.net/portals/0/documents/resources/gsr/2014/gsr2014_full%20report_low%20res.pdf

Application of Renewable Energy

- Renewable energy replaces conventional fuels in four distinct areas:
  - power generation
  - hot water/ space heating
  - transport fuels, and
  - rural (off-grid) energy services

Status of renewable energy technologies – characteristics and costs

Technology	Typical characteristics		Typical energy costs (US cents/kWh)
Large hydro	Plant size:	10 – 18,000MW	3-5
Small hydro	Plant size:	1 – 10MW	5-12
Biomass power	Plant size:	1 – 20 MW	5-12
Geothermal power	Plant size:	1 – 100MW	5-12
Rooftop solar PV	Peak capacity:	2 – 5 kW_{peak}	17-34
Utility-scale solar PV	Peak capacity:	200kW – 100MW	15-30
On-shore wind	Turbine size	1.5-3.5MW; rotor diameter 60-100m	5-9
Off-shore wind	Turbine size	1.5-5MW; rotor diameter 70-125m	10-20
Ethanol	Feedstocks	Sugarcane, sugar beets, corn, cassava, sorghum, wheat	30-50 cents/litre (sugar) 60-80 cents/litre (corn) (gasoline equivalent)
Biodiesel	Feedstocks:	Soy, rapeseed, mustard seed, palm, jatropha, waste vegetable oils and animal fats	40-80 cents/litre (corn) (diesel equivalent)
Biogas digester	Digester size:	6-8 m^3	n/a

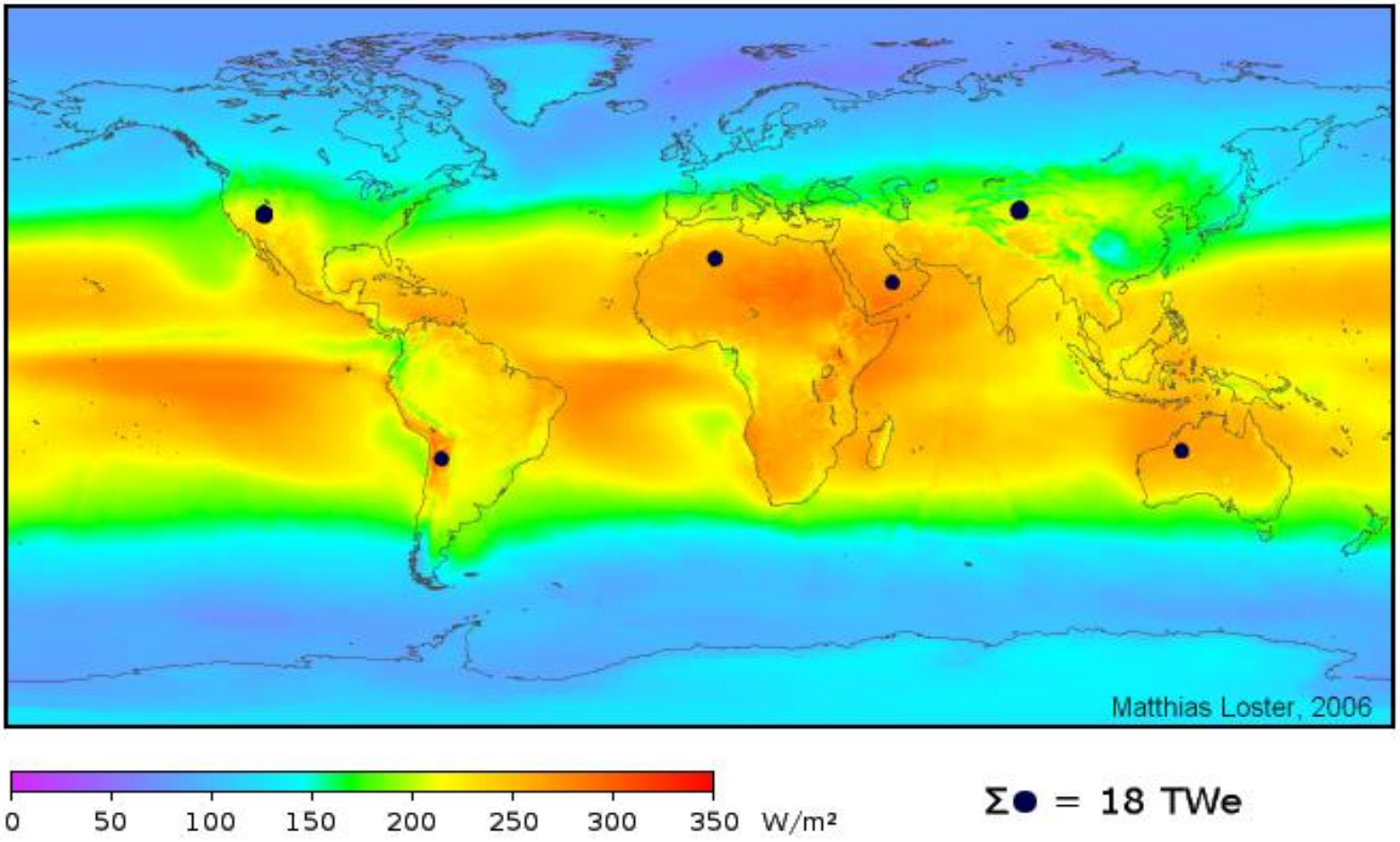
Solar energy

- Most renewable energy comes either directly or indirectly from the sun.
- Sunlight, or solar energy, can be used directly for heating and lighting homes and other buildings, for generating electricity, and for hot water heating, solar cooling, and a variety of commercial and industrial uses
- It has been used in many traditional technologies for centuries, and has come into widespread use where other sources of power are absent, such as in remote locations and in space.

Solar power

- **Solar power** is the conversion of sunlight into electricity, either directly using photovoltaics (PV), or indirectly using concentrated solar power (CSP).
- Concentrated solar power systems use lenses or mirrors and tracking systems to focus a large area of sunlight into a small beam.
- Photovoltaics converts light into electric current using the photoelectric effect.

- Solar areas defined by the dark disks could provide more than the world's total primary energy demand (assuming a conversion efficiency of 8%). That is, all energy currently consumed, including heat, electricity, fossil fuels, etc., would be produced in the form of electricity by solar cells.



Photovoltaics

- A solar cell, or photovoltaic cell (PV), is a device that converts light into electric current using the photoelectric effect.





Small solar PV





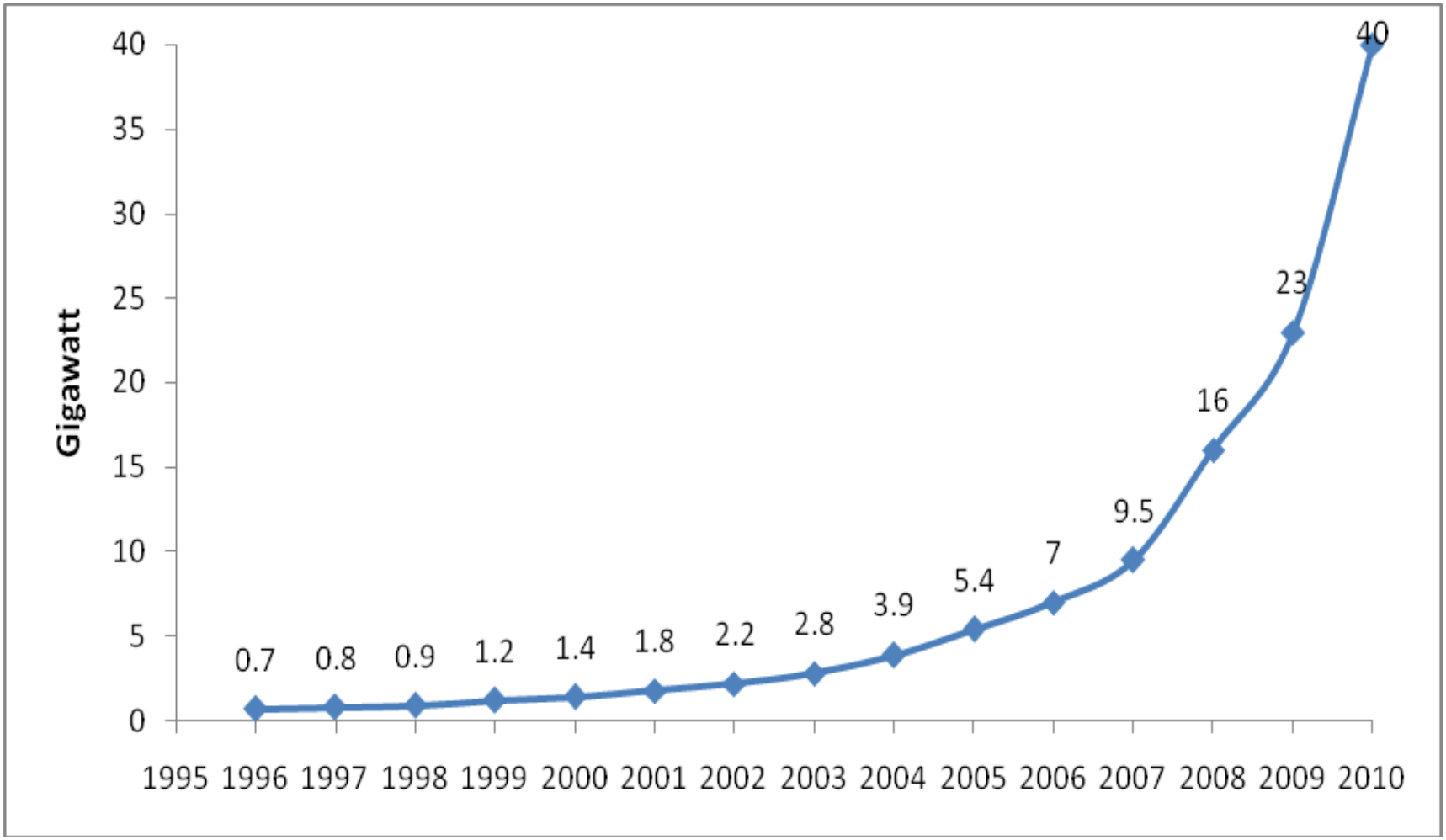
Concentrating Solar Power (CSP)

- CSP systems use lenses or mirrors and tracking systems to focus a large area of sunlight into a small beam.
- The concentrated heat is then used as a heat source for a conventional power plant.





Solar PV, Existing world capacity, 1995-2010



Wind Energy

- **Wind power** is the conversion of wind energy into a useful form of energy, such as using wind turbines to make electricity, wind mills for mechanical power, wind pumps for pumping water or drainage, or sails to propel ships
- Large-scale wind farms are connected to the electric power transmission network; smaller facilities are used to provide electricity to isolated locations.

Wind turbines



Hydropower

- **Hydropower** is power that is derived from the force or energy of moving water, which may be harnessed for useful purposes.
- **Hydroelectricity** is the term referring to electricity generated by hydropower; the production of electrical power through the use of the gravitational force of falling or flowing water. It is the most widely used form of renewable energy.
- It is estimated that Africa has 1,750 TWh potential of hydropower but only 5% of the continent's potential of hydropower has been exploited.

Hydropower dam







Extra Large Hydropower dam



Small hydropower







Biomass energy

- **Biomass** is biological material from living, or recently living organisms, such as wood, waste and alcohol fuels
- Biomass can be used to produce electricity, transportation fuels, or chemicals. The use of biomass for any of these purposes is called biomass energy
- The most conventional way in which biomass is used however, still relies on direct incineration.

Biomass











Biofuels

- **Biofuels** are a wide range of fuels which are in some way derived from biomass.
- Bioethanol is an alcohol made by fermenting the sugar components of plant materials and it is made mostly from sugar and starch crops.
- Biodiesel is made from vegetable oils, animal fats or recycled greases.
- Biogas is methane produced by the process of anaerobic digestion of organic material by anaerobes.

Bioethanol

- Ethanol is an alcohol-based fuel produced by fermenting plant sugars.
- It can be made from many agricultural products and food wastes if they contain sugar, starch, or cellulose, which can then be fermented and distilled into ethanol.



Biodiesel

- Biodiesel is the methyl or ethyl ester of the fatty acid made from virgin or used vegetable oils (both edible and nonedible) and animal fat.
- Biodiesel can be blended in any proportion with mineral diesel to create a biodiesel blend or can be used in its pure form.
- Just like petroleum diesel, biodiesel operates in the compression ignition (diesel) engine, and essentially requires very little or no engine modifications because the biodiesel has properties similar to mineral diesel.



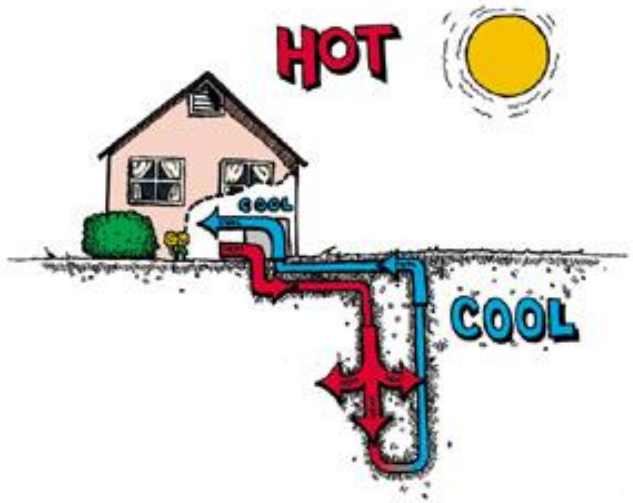


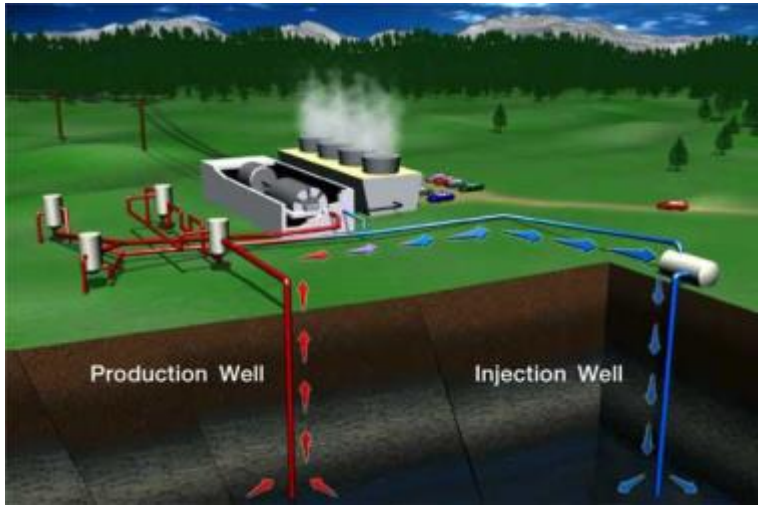
Geothermal energy

- **Geothermal energy** is thermal energy generated and stored in the Earth. Thermal energy is energy that determines the temperature of matter.
- Earth's geothermal energy originates from the original formation of the planet, from radioactive decay of minerals, from volcanic activity, and from solar energy absorbed at the surface.
- The geothermal gradient, which is the difference in temperature between the core of the planet and its surface, drives a continuous conduction of thermal energy in the form of heat from the core to the surface.

Geothermal

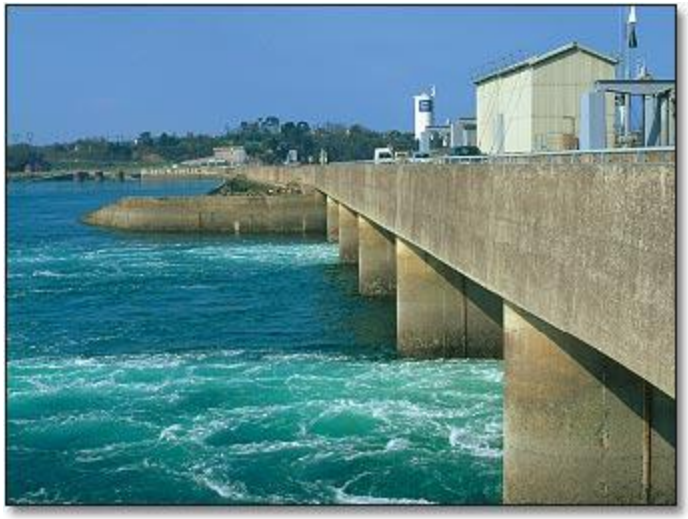
- Geothermal power is the use of geothermal heat to generate electricity.
- At the moment, Iceland and New Zealand are two of the largest users of geothermal energy, although many others also have potential.

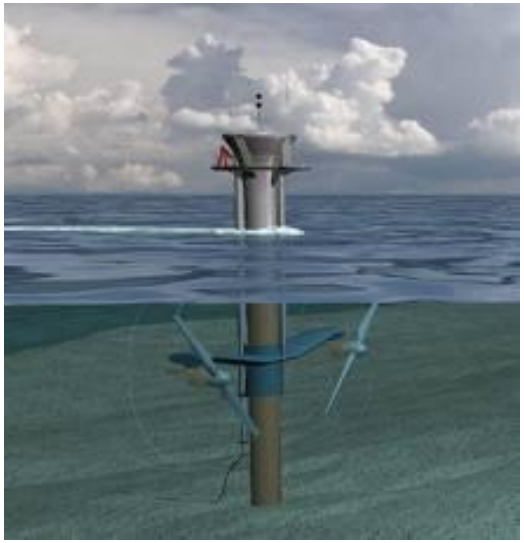




Tidal power

- Tidal power is the power achieved by capturing the energy contained in the moving water in tides and ocean currents.
- It is classified as a renewable energy source, because tides are caused by the orbital mechanics of the solar system and are considered inexhaustible.
- Geothermal power and tidal power are the only renewable energy sources not dependent on the sun but are today limited to special locations.





Ghana going tidal?

www.energymin.gov.gh/?p=3340

Apps | Free Spelling Lists - ... | New folder | New Tab | Home1 | Google Chrom | Other bookmarks

Ministry of Energy & Petroleum (MoEP), Ghana

ENERGY FOR A BETTER GHANA

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WORK BEGINS ON ADA WAVE POWER PROJECT

January 26, 2015 | Communications Unit

The first phase of power generation using energy from sea waves has started in Ada, in the Greater Accra region.

The project which is being undertaken by TC Energy is expected to contribute 14 megawatts of power to the national grid by next month.

The wave-powered project is considered environmentally friendly and will generate power which will cost less compared to Ghana's current power sources which include hydro and thermal energy.

The Chief Executive of TC Energy, Anthony Opoku said that the company is "looking at about 14,400 kilowatt hour per day that will come from the six wave converter units that we are going to deploy under the sea".

He lauded Ada as the perfect coastal town for such a project, enthusing that the Ada coast has unique waves which will be of huge benefit to the project.

"This technology, most of it depends on the wave height and that's why we won't be worried too much about

WELCOME

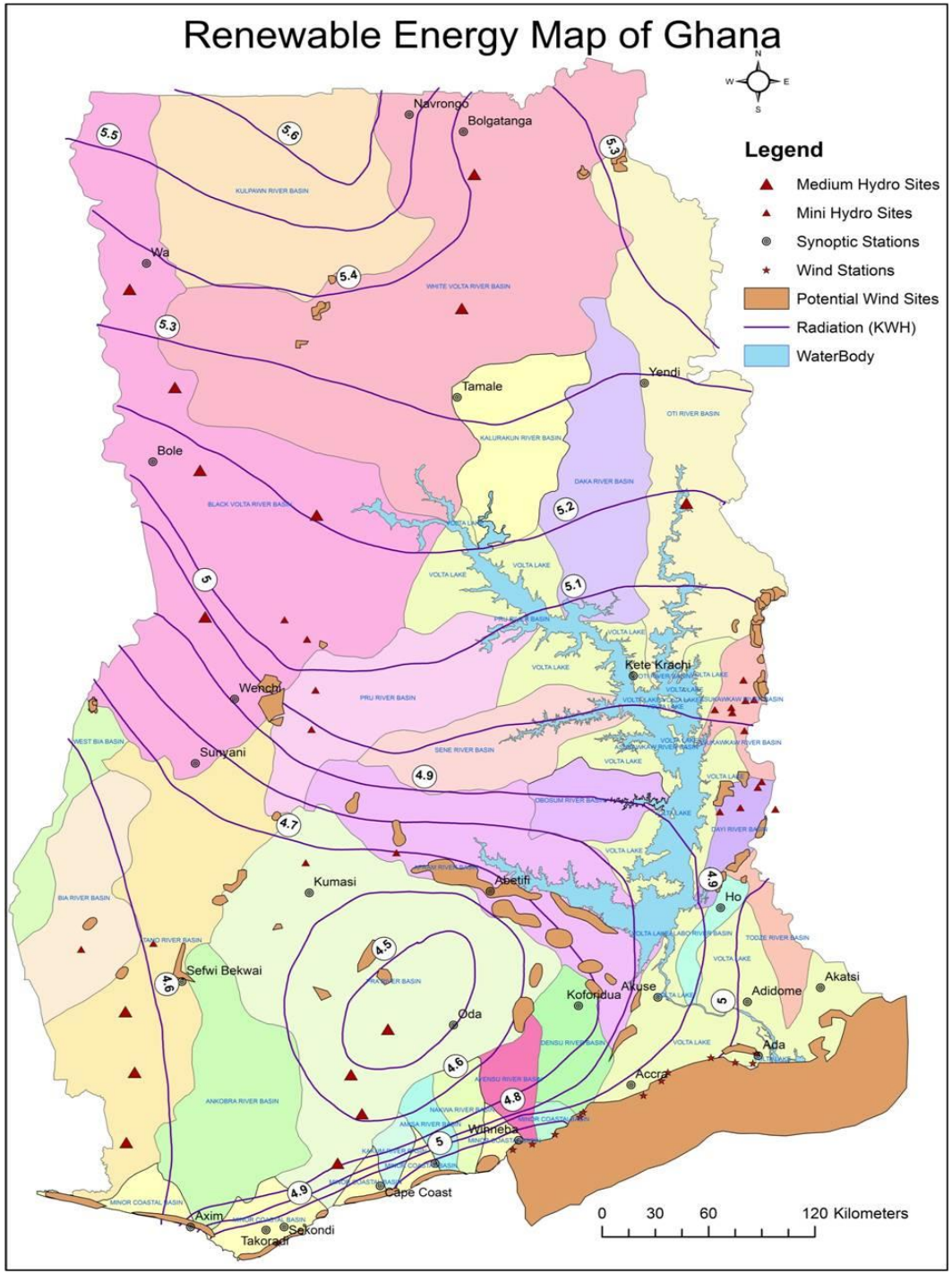


Hon. Emmanuel Armah-Kofi Buah
Energy & Petroleum Minister
MP for Ellembele

04:49
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Renewable Energy in Ghana

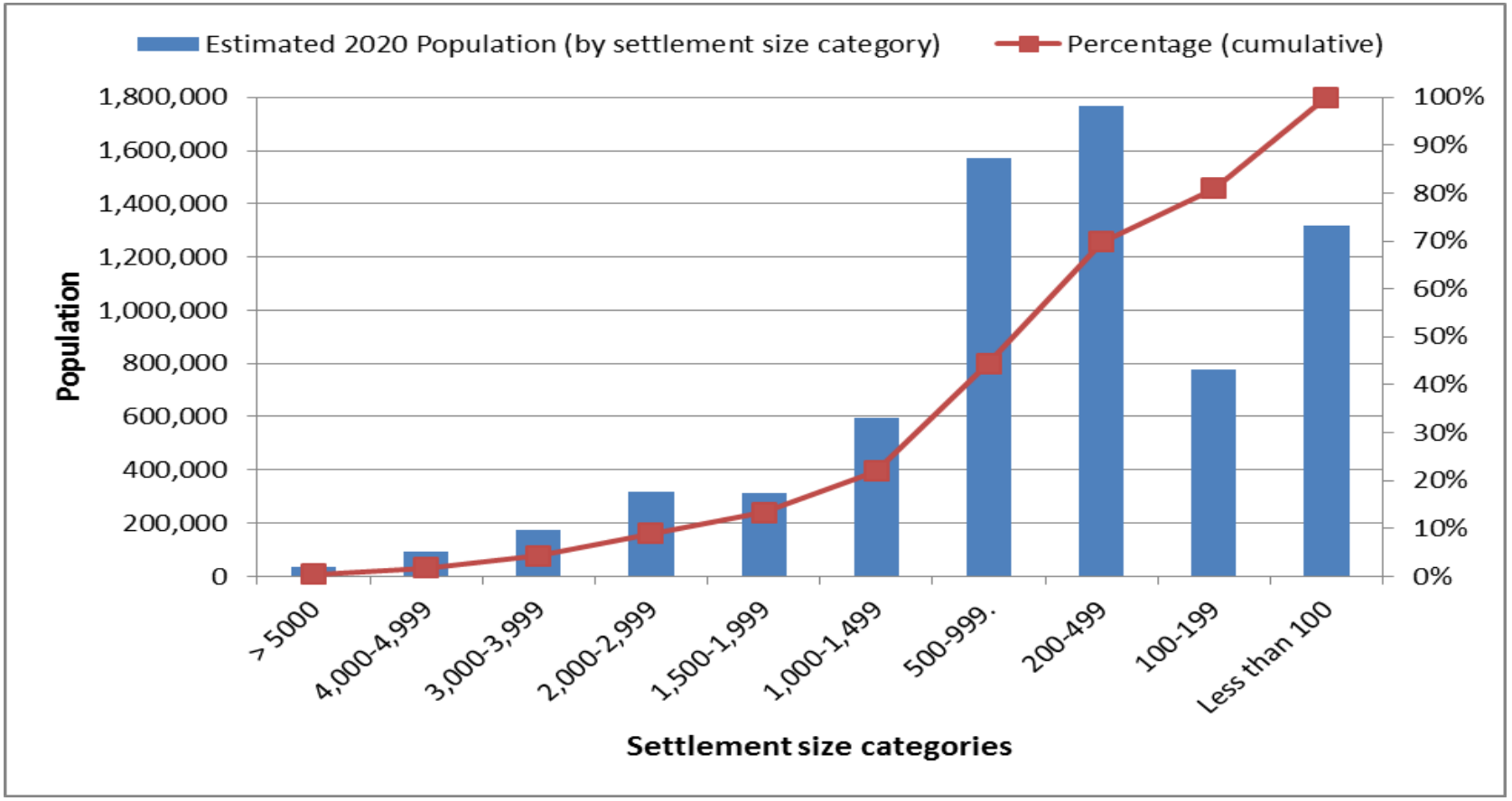
- Great potential but little development
  - Solar
  - Wind
  - Biomass
  - Biofuels
  - Hydro


Renewable Energy Map of Ghana
N
W
E
S
Legend
Medium Hydro Sites
Mini Hydro Sites
Synoptic Stations
Wind Stations
Potential Wind Sites
Radiation (KWH)
WaterBody
Navrongo
Bolgatanga
Wa
Tamale
Yendi
Bole
Wenchi
Sunyani
Kete Krachi
Kumasi
Abetifi
Ho
Sefwi Bekwai
Koforidua
Akuse
Adidome
Akatsi
Oda
Ada
Accra
Winneba
Cape Coast
Axim
Sekondi
Takoradi
5.5
5.6
5.3
5.4
5.3
5
5.2
5.1
4.9
4.7
4.9
4.5
4.6
4.6
4.8
5
4.9
5
0
30
60
120 Kilometers

National Electrification Scheme (NES)

- 30-yr Electrification Plan developed in 1989 to achieve universal access to electricity for communities with population over 500 by 2020
- NES review was conducted in 2010
  - To evaluate the strategy and approach adopted so far and make recommendations for the effective pursuit and achievement of the NES goals
  - To prepare a revised Master Plan for achieving 100% access by 2020
- **Revised plan recommended grid electrification** to 2643 communities (above 500 inhabitants in 2010 or 500 by 2020) at a total cost of GHS 886 million

Challenges and gaps



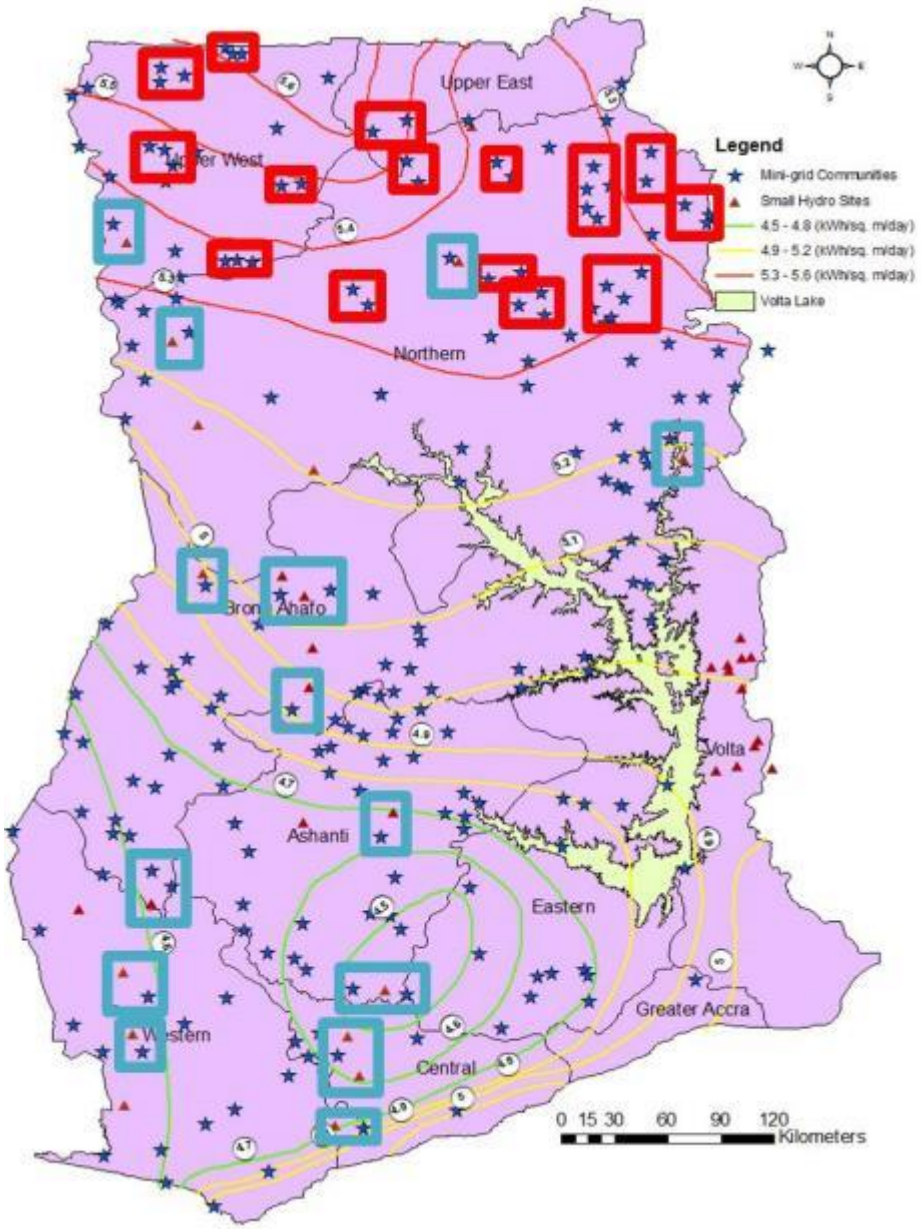
Population projection shows that about 3 million people could be living in communities with population less than 500 in 2020. What happens to the majority of these communities who are unlikely to be covered by grid electrification by 2020?

Network Planner Computer Model Results by Region

Region	Number of un-electrified communities	Total population of un-electrified communities	Percentage of un-electrified communities for which each electrification technology is recommended			Total cost of off-grid (US$) (10yr, initial + recurring)		Total cost of mini-grid (US$) (10yr, initial + recurring)		Total cost of grid (US$) (10yr, initial + recurring)	
			Off-Grid	Mini-Grid	Grid	Regional cost	Per HH	Regional cost	Per HH	Regional cost	Per HH
Ashanti	221	143,838	5%	15%	81%	3,120,000	3,560	8,320,000	3,220	49,700,000	2,210
Brong Ahafo	195	154,134	4%	21%	75%	2,560,000	3,560	10,300,000	3,260	48,400,000	2,150
Central	175	114,722	1%	3%	96%	401,000	3,430	2,230,000	3,100	54,500,000	1,890
Eastern	247	125,202	2%	6%	92%	1,700,000	3,600	3,540,000	3,190	44,600,000	2,120
Greater Accra	11	7,160	-	9%	91%	-	-	439,000	3,020	2,420,000	1,760
Northern	660	400,772	20%	10%	70%	28,500,000	3,490	16,500,000	3,300	102,000,000	2,400
Upper East	299	210,309	1%	1%	98%	706,000	3,400	939,000	3,110	60,000,000	1,880
Upper West	294	200,496	10%	9%	81%	7,710,000	3,400	7,010,000	3,230	54,000,000	2,250
Volta	179	182,915	2%	2%	96%	1,840,000	3,440	2,370,000	3,170	80,000,000	2,030
Western	319	224,803	-	6%	94%	-	-	6,000,000	3,330	95,600,000	2,130
National	2600	1,764,351	7%	8%	85%	46,537,000	3,480	57,648,000	3,190	591,220,000	2,080

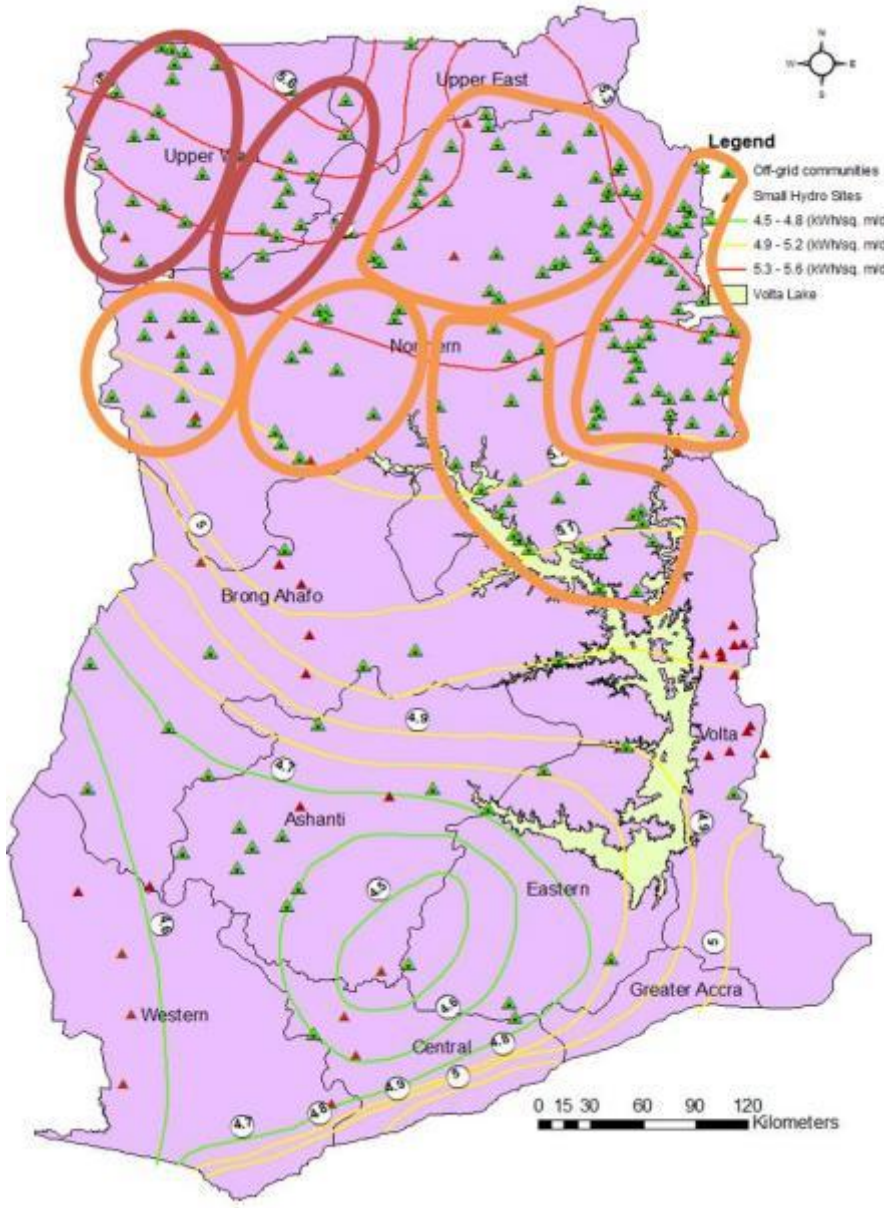
Map of mini-grid communities showing solar radiation levels and small hydro resources

- Solar mini-grids in Upper and Northern Regions should be considered
- Many cases of mini-grid communities close to potential small hydro sites in Upper West, Northern, Brong Ahafo, Ashanti and Western Regions



Map of off-grid communities showing solar radiation levels and small hydro resources

- Many off-grid clusters in Upper West and Northern Regions, including some of the areas where solar radiation in Ghana is highest, between 5.2 and 5.6 kWh/m2/day



END OF LECTURE 2

QUESTIONS